

NYC Taxi Trip Duration Analysis

Machine Learning Analysis of NYC Yellow Taxi Trip Data

Springboard Capstone 3
Ben Chretien | May 10, 2026

Problem Identification

- Predict NYC taxi trip duration
- Identify key factors influencing travel time
- Improve transportation planning and routing efficiency

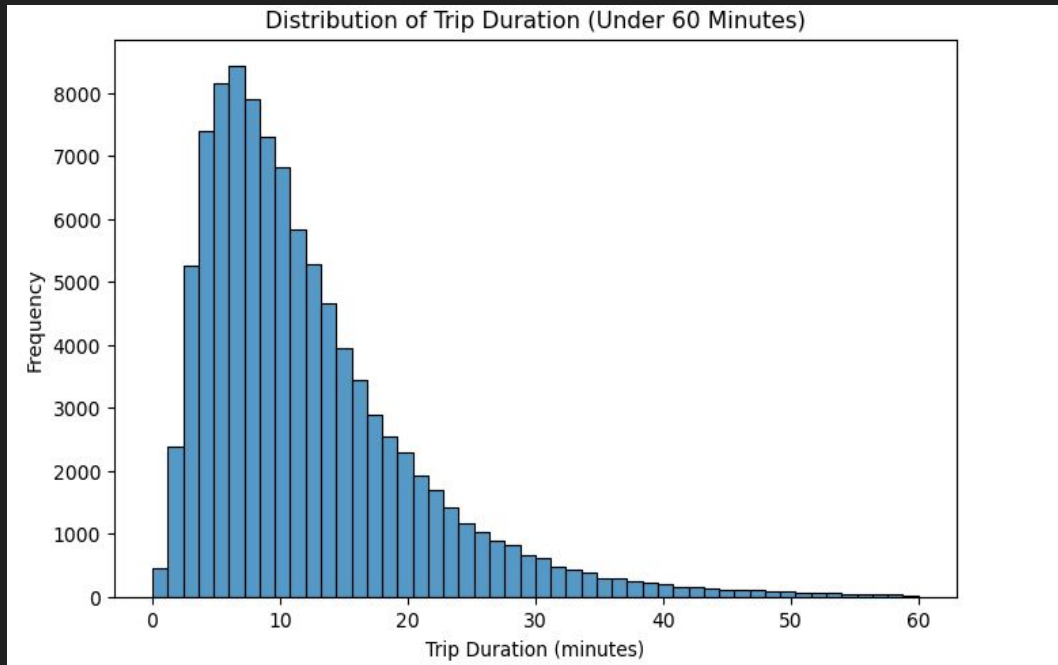
Data

- NYC Yellow Taxi trip-level dataset
- Includes pickup/dropoff timestamps, trip distance, passenger count, fare, and location data
- Dataset contains transportation activity across New York City
- Used trip duration as the target variable for prediction

Data Preparation / Preprocessing

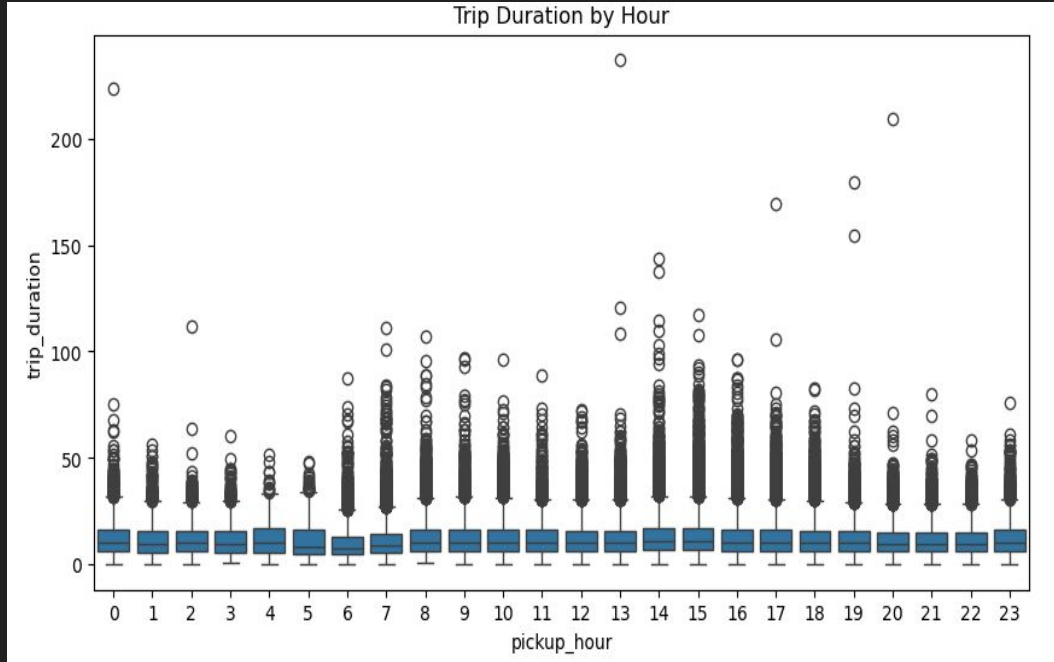
- Converted timestamps to datetime format
- Created trip duration target variable
- Removed invalid and extreme trip durations
- Engineered pickup hour and pickup day features
- Prepared training and testing datasets

EDA: Trip Duration Distribution



- Most taxi trips were relatively short in duration
- Distribution was heavily right-skewed
- Extreme outliers were identified and removed during preprocessing

EDA: Average Trip Duration by Pickup Hour



- Trip duration varied throughout the day
- Longer trips often occurred during commuting periods
- Traffic patterns likely influenced travel time variability

Modeling Approach

- Split data into training and testing sets
- Applied feature scaling for linear models
- Trained Linear Regression and Ridge Regression models
- Used Random Forest Regression for non-linear modeling
- Evaluated models using R^2 and RMSE

Models Used

- Linear Regression
- Ridge Regression
- Random Forest
 - Refined Random Forest

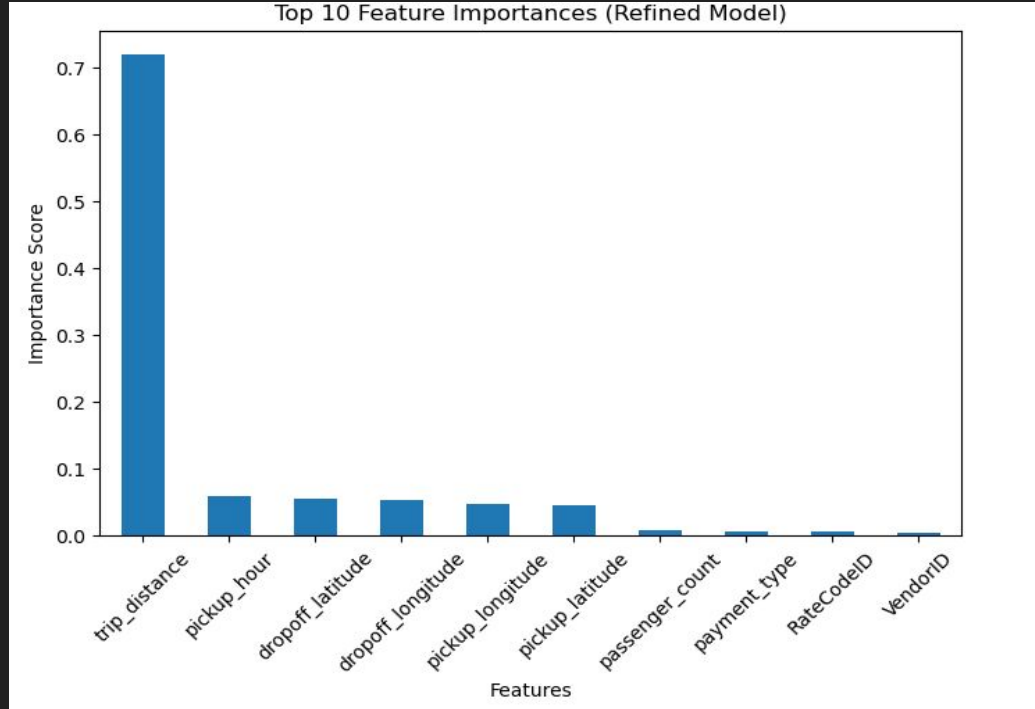
Model Results

- Linear Regression: R^2 : 0.71 | RMSE: 5.00
- Ridge Regression: R^2 : 0.71 | RMSE: 5.00
- Random Forest: R^2 : 0.88 | RMSE: 3.29
- Refined Random Forest: R^2 : 0.74 | RMSE: 4.80

Final Model Selection

- Original Random Forest achieved the strongest predictive performance
- Feature importance analysis showed strong reliance on fare-related variables
- Refined Random Forest removed fare-related variables for improved interpretability
- Trip distance emerged as the strongest independent predictor of trip duration

Feature Importance Analysis



- Trip distance emerged as the strongest independent predictor
- Geographic and time-based features also contributed to travel time variability

Recommendations

- Use trip distance and time-based patterns to improve travel time estimation
- Incorporate traffic and congestion trends into routing analysis
- Continue refining feature selection to improve model interpretability

Conclusion / Future Work

Conclusion

- Trip duration influenced by multiple interacting factors
- Trip distance emerged as the strongest independent predictor
- Random Forest models captured complex travel patterns effectively

Future Work

- Incorporate weather and traffic data
- Explore advanced boosting models such as XGBoost
- Perform additional feature engineering and hyperparameter tuning